

Anchor Facts: Twos, Fives, and Tens

Answer Key

Grade 3

Quick Answers

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|--|---|
| 1. 4th number = 20, 5th number = 25, 6th number = 30 | 5. $5 \times 8 = 40$, $6 \times 8 = 48$ |
| 2. $2 \times 8 = 16$, $8 + 8 = 16$ | 6. $10 \times 6 = 60$, $9 \times 6 = 54$ |
| 3. $10 \times 4 = 40$, $10 \times 8 = 80$ | 7. $9 \times 8 = 72$, — |
| 4. $2 \times 6 = 12$, $4 \times 6 = 24$ | 8. $7 \times 6 = 42$, — |
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What is verified

6 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 3

3.OA.C.7 – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–4 of 5 across 17 tagged checks.

1. Count by fives: 5, 10, 15, __, __, __

Solution: Each step adds 5 to the number before it: $15 + 5 = 20$, then $20 + 5 = 25$, then $25 + 5 = 30$. The 6th number, 30, is 5×6 .

20, 25, 30

2. Array with 2 rows of 8: 2×8 and $8 + 8$.

Solution: The two rows hold 8 dots each, so $2 \times 8 = 16$ — and adding the rows, $8 + 8 = 16$. A twos fact *is* a double: same array, two names.

16 and 16

3. 10×4 and 10×8 .

Solution: Counting by tens: four tens are 40, eight tens are 80. $10 \times 4 = 40$ and $10 \times 8 = 80$.

40 and 80

4. 2×6 , then double it for 4×6 .

Solution: The anchor: $2 \times 6 = 12$. Four groups is twice two groups, so double it: $12 + 12 = 24$. So $4 \times 6 = 24$. 12 and 24

5. 5×8 , then one more 8 for 6×8 .

Solution: The anchor: $5 \times 8 = 40$. Six groups is one more group of 8 than five groups: $40 + 8 = 48$. So $6 \times 8 = 48$. 40 and 48

6. 10×6 , then one group of 6 away for 9×6 .

Solution: The anchor: $10 \times 6 = 60$. Nine groups is one group of 6 fewer than ten groups: $60 - 6 = 54$. So $9 \times 6 = 54$. 60 and 54

7. Choose an anchor family for 9×8 , then use it.

Solution: Tens is the most direct choice: $10 \times 8 = 80$, and nine groups is one 8 fewer, $80 - 8 = 72$. (Fives also works: $5 \times 8 = 40$ and $4 \times 8 = 32$, and $40 + 32 = 72$ — credit any anchor that is actually used.) 72

8. Challenge: use any anchor fact for 7×6 .

Solution: The fives anchor is closest: $5 \times 6 = 30$, and the two more groups are $2 \times 6 = 12$, so $30 + 12 = 42$. A correct explanation names the anchor fact and says how it built the answer (e.g. “I knew five 6s, then added two more 6s”). Other anchors count too: doubling 2×6 twice and adding one more 6, or $10 \times 6 = 60$ minus three 6s. 42